

Fitness drivers of division of labor in vertebrates

Reviewed Preprint

v1 • May 13, 2025

Not revised

Irene García-Ruiz , Dustin Rubenstein

Department of Ecology, Evolution and Environmental Biology, Columbia University, New York, United States

 https://en.wikipedia.org/wiki/Open_access Copyright information

eLife Assessment

This **valuable** work formulates an individual-based model to understand the evolution of division of labor in vertebrates, in particular, to examine the role of indirect versus direct fitness benefits. The evidence supporting the main conclusions is **incomplete** at this stage, with key details of simulation assumptions not adequately described and exploration of alternative assumptions and parameter space lacking.

<https://doi.org/10.7554/eLife.105501.1.sa3>

Abstract

Although division of labor as a means to increase productivity is a common feature in animal social groups, most previous studies have focused almost exclusively on eusocial insects with extreme task partitioning. Empirical evidence of division of labor in vertebrates is scarce, largely because we lack a theoretical framework to explore the conditions under which division of labor is likely to evolve. By explicitly considering alternative helping tasks with varying fitness costs, we model how individual decisions on task specialization may influence the emergence of division of labor under both direct and indirect fitness benefits. Surprisingly, we find that direct survival benefits of living in larger groups are the primary force driving the evolution of cooperation to enhance group productivity, and that indirect fitness benefits derived from related group members are only a non-essential facilitator of more stable forms of division of labor. In addition, we find that division of labor is favored by increasingly harsh environments. Ultimately, our model not only makes key predictions that are consistent with existing empirical data, but also proposes novel avenues for new empirical work in vertebrate and invertebrate systems alike.

Introduction

Division of labor as a means to boost productivity holds significance not only in human societies, but also those of other animals [1 –5 ]. Nearly 250 years ago, the economist Adam Smith [2 ] proposed that division of labor via task specialization within human societies not only improves individual efficiency and group productivity, but also reduces the costs associated with switching

tasks. Similarly, in nonhuman animal societies, division of labor to increase group reproductive output (analogous to economic gain) is characterized by group members specializing in particular tasks, either temporarily or permanently [6–8]. To date, all theoretical and the vast majority of empirical work on the selective pressures favoring division of labor via task specialization in animal societies has focused on eusocial insect societies characterized by sterile workers that are unable to ever reproduce [8–12]. Since workers in eusocial societies can, in most cases, only obtain fitness benefits indirectly through kin selection by maximizing their colony reproductive output, selection can occur at the colony level, typically resulting in very low conflict of interest between group members. However, complex societies occur outside of eusocial insects, such as in many birds and mammals, including humans. Because sterile workers are absent in vertebrate societies, conflict can be much higher than in eusocial insect societies. The few studies of division of labor outside of eusocial insects have focused exclusively on reproductive division of labor (i.e., reproductive skew), despite the fact that temporal or permanent non-reproductive organisms can engage in an array of different helping tasks [3, 13]. Given the scarcity of studies examining the selective advantages of division of labor beyond eusocial insects (but see [14]), our understanding of the evolutionary pressures of division of labor is largely confined to instances where task specialization is linked to sterility and where conflict of interest is low.

In cooperatively breeding vertebrates, which live in societies with more equitable sharing of reproduction (i.e., low reproductive skew), low or mixed kinship, and often have unrelated helpers, conflict of interest can be high and direct fitness benefits may be more relevant in the evolution of division of labor than indirect benefits [15–17]. In fact, shared fitness incentives to increase group productivity are not limited to eusocial insects with high within-group relatedness and sterile workers. In cases where group size positively correlates with an increase in the survival or reproduction of group members, division of labor via task specialization may also evolve under direct fitness benefits to subordinate members of a group (i.e., group augmentation [18, 19]). Although a great deal of recent empirical work highlights the importance of direct benefits in the evolution of cooperative breeding behavior in vertebrates [20–23], the joint influence of direct and indirect fitness benefits in the evolution of division of labor has, to our knowledge, never been investigated.

Most previous models of division of labor via task specialization considered all helping tasks under the same fitness umbrella, which is a good approximation in eusocial insect societies because colony members share fitness costs and benefits. In vertebrates, however, helpers may obtain fitness benefits directly via reproduction, and thus, selection occurs more strongly at the individual level than it does in most eusocial insects. Group members in vertebrate societies may then contribute to distinct helping tasks differently according to individual life history strategies [24–26] or differences in the magnitude or type of fitness costs and benefits that they receive from cooperative tasks [27]. One of the key distinctions between fitness costs is whether a task impacts more immediate survival or direct reproduction. For instance, defensive behaviors such as predator mobbing may have a high risk of injury and death [28–30], whereas other activities such as allofeeding or nest/territory maintenance may affect body condition owing to lower feeding rates and time and energy investment, consequently affecting their own reproductive output [31–33]. Therefore, changes in life histories, such as those linked to the likelihood of becoming a breeder, may select for task specialization according to the relative direct fitness cost associated with different helping tasks.

In addition to life history differences, demographic, social, and ecological factors that affect help workload and the prospect of helpers becoming breeders are likely to have a strong impact on task allocation because they alter the likelihood of obtaining direct fitness benefits [34, 35]. For example, age is known to be an important factor influencing helping workload in vertebrates [36, 37], is often correlated with differences in other life-history traits [33, 38–40], and may be a proxy for other processes involved in the evolution of help [41, 42]. Moreover, age-dependent task allocation (i.e., temporal polyethism) is common in eusocial insects that do not

show morphological differentiation [43]. Social and ecological factors are also thought to influence the emergence of division of labor in eusocial insects [44–46]. Indeed, high relatedness may reduce conflict of interest between group members and promote the evolution of division of labor [47–51]. Similarly, in harsh environments where group size strongly influences survival [52–55], division of labor may be favored to more efficiently increase group productivity.

To disentangle the role of direct versus indirect fitness benefits in the evolution of division of labor in vertebrate societies, we created a stochastic model in which we varied the influence of different evolutionary forces to understand under which social and ecological circumstances division of labor may arise to maximize the inclusive fitness of individuals in a group. We used distinct tasks with different fitness costs to maximize the group reproductive output when they were performed to a similar extent. Individuals in our model were allowed to evolve different behavioral responses or reaction norms in task specialization, depending on their likelihood of becoming breeders. We compared the results of this model to those of a benchmark model in which the reproductive outcome was independent of the type of task performed by group members. We present the results for different environmental conditions in the presence or absence of relatedness (indirect) or group size (direct) benefits. Ultimately, this work demonstrates that the direct fitness benefits of group living attained in increasingly harsh environments are key to the evolution of division of labor in vertebrates.

Results

Rationale

Our model consists of a population characterized by overlapping generations residing in a habitat featuring a limited number of breeding territories. These territories are monopolized by groups composed of a single breeder and an unspecified number of subordinate helpers, each of which is capable of reproducing during its lifetime. Individuals in the population fall into one of three categories: (1) *breeders* that monopolize reproduction; (2) *subordinates* that may perform different helping tasks, either in their natal group or in an outside group to which they dispersed; and (3) *floaters* that dispersed from their natal group but did not join another group.

We assume complete reproductive skew such that subordinate group members are prevented from reproducing at a given timestep and may only help raise the offspring of the dominant breeders. Dispersers cannot reproduce without acquiring a territory (i.e., floaters). Helpers are not sterile and may inherit the breeding position when the breeder dies. When this occurs, helpers and dispersers from other groups compete for the breeding position and win with a probability proportional to their dominance value (Table 1). We use the term dominance value to designate the competitiveness of an individual compared to other candidates in becoming a breeder, regardless of group membership, considering both age as a proxy for resource holding potential and helping costs to body condition.

Subordinate helpers in our model may choose between two distinct tasks with differing fitness costs: (1) *defensive tasks* (i.e., defense of the dominant breeders' offspring at cost to their own survival); or (2) *work tasks* (i.e., tasks with an associated cost to their body condition and consequently to their competitiveness to become breeders in the future). Subordinate helpers can then evolve different workloads and exhibit task specialization that is either fixed throughout their lifetime or varies with their dominance value. The parameters x_h and y_h (Table 1) reflect how much the subordinates' survival and dominance values decrease with increasing investment in defense or work tasks, respectively. Although we do not assume an inherent difference in body condition between individuals when born, variance in age and helping propensities can create differences in the fitness costs linked to task performance.

Symbol	Meaning	Value
<i>Genes</i>		
α	Genetic propensity to help	0
β	Genetic predisposition to disperse	1
γ_0	Intercept in the task specialization reaction norm	0
γ_t	Effect size of age on task specialization	0
<i>Scaling parameters</i>		
y_h	Effect size of cost of help on dominance when performing “work tasks”	0.1
x_h	Effect size of cost of help in survival when performing “defensive tasks”	3 / 5 / 7
x_n	Effect size of the benefit of group size in survival	0 / 3
x_0	Intercept in the survival function	1.5*
m	Baseline mortality	0.1 - 0.3
f	Mean number of groups a floater samples for becoming a breeder	2*
k_m	Deviation from a perfect split in the need of both defense and work tasks	0.1
k_h	Effect size of the cumulative help of subordinates on the fecundity of the breeder	1
k_0	Fecundity of the breeder in the absence of help	1
t_{min}	Minimum age to come breeder in the scramble context implementation	2
q	Baseline individual quality in the scramble context implementation	4
μ	Mutation rate	0.05
σ_μ	Mutation step size	0.04
N_b	Number of breeding territories	5000
<i>Phenotypic traits</i>		
H	Level of help of either kind provided to the breeder $H = \alpha$. We express cumulative level of help given in a group as $\sum H_i$	
D	Dispersal propensity $D = \beta$, applying boundaries between 0 and 1	
R	Dominance value	
T	Probability of choosing “defensive task” vs “work task”	
S	Survival rate	
K	Fecundity of the breeder	
<i>Variables</i>		
t	Age	
N	Emergent group size	
N_f	Total number of floaters	
$N_{f,b}$	Number of floaters bidding for a breeding position, given by $N_{f,b} = f * N_f / N_b$	

Table 1.

Overview of notation.

Values conveyed for the genes are initial input values, values given for the scaling parameters are fixed throughout the simulations. If more than one value is given, results are shown to display the effect of this parameter’s variation. *: Additional values of these parameters are discussed in the Supplemental Material.

We assume net survival benefits from living in large groups (i.e., group augmentation), which occurs in many social species [53, 54, 56, 57]. Subordinate members of a group may therefore gain both direct and indirect fitness benefits from helping to increase the productivity of breeders, if doing so results in an increase in group size. In the model, the benefit of alloparental care to the productivity of the breeder is maximized when group members provide overall similar workloads for both defensive and work tasks, since both are often needed for the survival of the offspring [58, 59]. To compare between the helper's preferred task and the task needed to maximize group productivity, we created a benchmark model in which the breeder's productivity is not constrained by the type of help provided. Higher productivity translates to larger group sizes, which enhances the survival prospects of all group members. Hence, subordinates have an incentive to develop division of labor. Subordinates that stay in their natal group may also gain improved fitness by indirectly passing on their genes through kin. Both group size and relatedness in our model are emergent properties of a series of demographic processes, including mortality, dispersal decisions, and help-dependent breeder fertility.

To distinguish between the effects of group augmentation and kin selection on inclusive fitness, we created two additional models. In the first model, we removed the benefits of group size (group augmentation) by setting the parameter $x_n = 0$ (Table 1). In the second model, we removed the effect of kin selection by simulating a cross-foster manipulation in which offspring just born are interchanged between different groups (see Methods for additional details).

To assess how habitat quality affects task specialization and the emergence of division of labor, we systematically varied habitat quality across simulations. Harsh environments are those that have high mortality (i.e., $m = 0.3$; Table 1), which also raises the turnover of the breeding positions. Lower values of m reflect an increased likelihood of individuals surviving another life cycle and, therefore, the chances of habitat saturation. The steps in the lifecycle of the model are illustrated in Figure 1.

Direct vs indirect fitness benefits

We first examined how both direct fitness benefits from group living and indirect fitness benefits from aiding kin influence the evolution of division of labor through task specialization across environments of varying quality. Our results show that division of labor initially emerges because of direct fitness benefits (Figure 2; GA), but a combination of direct and indirect fitness benefits leads to higher rates and more stable forms of division of labor (Figure 2; KS + GA). It is important to note that, as depicted in Figure 2, intermediate values of task specialization indicate in all cases age/dominance-mediated task specialization ($y_t \neq 0$; Table 1) and never a lack of specialization ($y_t = 0$; Table 1); further details are shown in the next section “*The role of dominance in task specialization*”.

In contrast to expectations from models of eusocial insects, our vertebrate model does not readily evolve division of labor when only kin selection is considered (see Supplemental Material for more details). Specifically, forms of help that impact survival never evolve under any environmental condition when only kin selection occurs (Figure 2; KS). This occurs because with kin competition for the breeding position and no survival benefits of group membership, the reduced incentives to remain in the natal group drive most subordinates to disperse and breed elsewhere (average dispersal rate between 88% in harsh environments and 93% in benign environments; Table S1). The few subordinates that remain in the natal territory help to obtain indirect benefits (average within-group relatedness between 0.36 in benign environments and 0.72 in harsh environments; Table S1). Even though individuals in our model cannot adjust their help according to their relatedness to the breeder, the high levels of relatedness that evolved under this paradigm indicate that allowing for targeted help would not qualitatively change the results. Survival costs of defensive tasks are then avoided, as mortality without group benefits is relatively high, while costs to dominance value are small for close relatives because of indirect

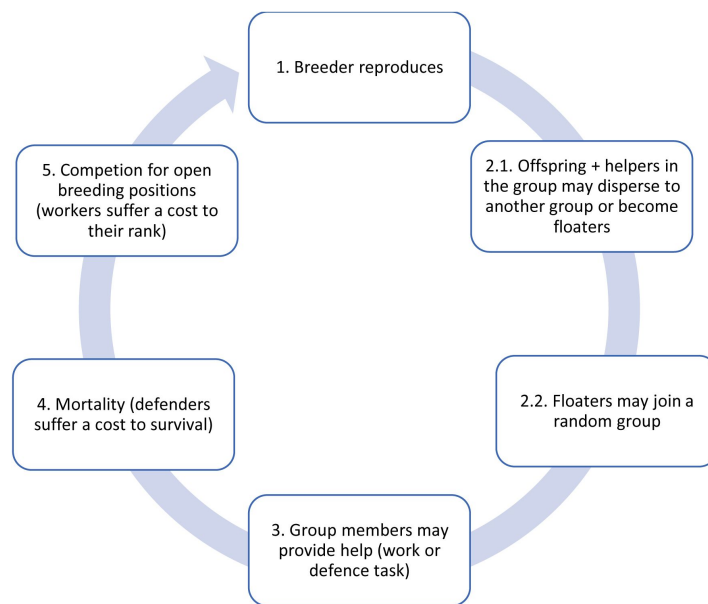


Figure 1.

Diagram of the scheduling executed per generation.

(1) A breeder reproduces. Its productivity depends on the cumulative level of brood care provided by the group during the previous life cycle. Maximum productivity is achieved when different helping tasks are performed to a similar extent. (2) Subordinates may disperse to become floaters, or they may stay in the group and help. Dispersers/floaters may join a random group to become subordinates. (3) Subordinates in the group either work to provision to the breeder's offspring or display defensive forms of help. (4) Individuals survive contingent on group-living benefits and dispersal costs, as well as the cost of defensive activities. (5) If a breeder dies, helpers in the group and a sample of floaters compete for the breeding position. Individuals still alive ascend one age class, and the cycle starts all over (i.e., next generation).

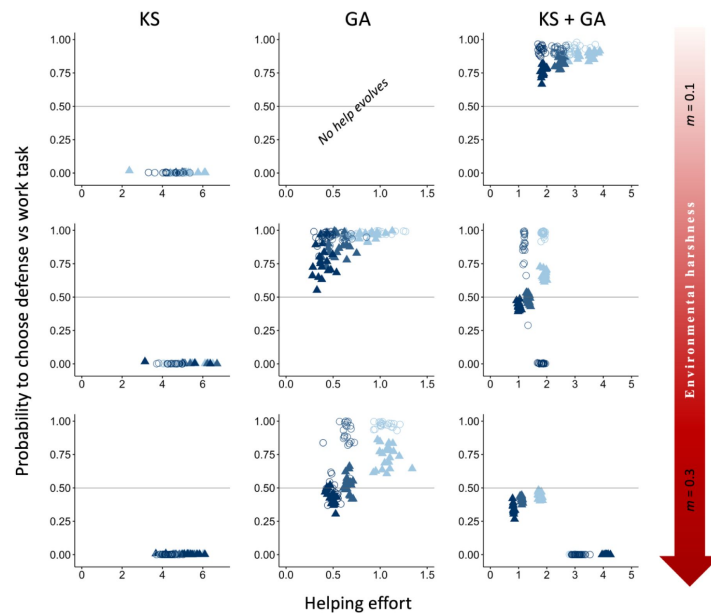


Figure 2.

Effect of environmental quality on alloparental care and division of labor.

The evolutionary equilibria for levels of helping and task specialization are shown at three different levels of environmental quality, ranging from benign ($m=0.1$) to harsh ($m=0.3$), and for three different levels of cost of help on survival (light blue, $x_h=3$; blue, $x_h=5$; and dark blue, $x_h=7$), across 20 replicas. The y- axis expresses the likelihood of individuals choosing a defensive task with a cost to survival versus a work task with a cost to their dominance rank. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor ($\blacktriangle$) or when no restrictions were introduced to the task performed by the group members ($\circ$). In each environment, additional details are given on the selective forces that play a role in the evolution of help and task specialization: help can only evolve by kin selection (KS), group augmentation (GA), or both (KS + GA). All parameter values described in [Table 1](#).

fitness benefits and low conflict of interest (**Figure 2**; KS). Adjusting input parameters to obtain higher survival rates and favoring philopatry did not qualitatively change the results, except for extreme values of baseline survival (see Supplementary Material), which indicates that kin selection alone is unlikely to select for the evolution of defensive tasks and division of labor in vertebrates.

Conversely, under only direct fitness benefits (group augmentation), tasks associated with costs to the dominance value are costlier because the only path to obtain fitness benefits is via direct reproduction. Therefore, defensive tasks that do not affect the dominance value are favored (**Figure 2**; GA). Under this scenario, division of labor can evolve to maximize the reproductive output of the breeders (**Figure 2**; GA, $m = 0.3$, $x_h \leq 5$), but also because it is the optimal strategy (**Figure 2**; GA, $m = 0.3$, $x_h = 7$). In benign environments, however, no help evolves due to the lower incentives to increase survival and the steep competition for reproduction due to the resulting habitat saturation.

When subordinates can obtain both direct and indirect fitness, division of labor is more strongly selected in increasingly harsher environments to increase breeder productivity (**Figure 2**; KS + GA). Overall, in moderate and harsh environments, work tasks with costs to the dominance value are more strongly selected than in benign environments due to high mortality increasing both the breeder's turnover, reducing breeding competition (lower effective costs of work tasks), and the costs to survival (high costs of defensive tasks). However, in more benign (and often highly productive) environments with habitat saturation, help can only evolve initially under kin selection and defensive tasks are favored because competition for the breeding position is lower in family groups. In this scenario, kin selection is achieved not only by subordinates in their natal group increasing the productivity of a related breeder (**Figure 2**; KS), but also by the increased survival of their siblings when augmenting the group size (**Figure 2**; KS + GA), regardless of whether they are related to the current breeder.

Overall, division of labor seems to be more likely to evolve in increasingly harsh environments where raising breeder productivity effectively enhances the survival of both helpers and relatives. In contrast, help evolves in benign environments, largely due to kin selection benefits and subordinates performing their preferred task of defending the breeder's offspring.

The role of dominance in task specialization

In the previous section, we explored the role of direct and indirect fitness benefits in the evolution of division of labor via task specialization. However, determining whether parameter γ_t (i.e., the slope of the reaction norm for the dominance value in task specialization) significantly differs from zero is crucial. Intermediate values of phenotypic task specialization may indicate division of labor among different age- or dominance-related groups (if $\gamma_t \neq 0$), or lack of specialization in either task (if $\gamma_t = 0$). Hence, for age-polyethism or temporal division of labor to arise in our model, individuals must switch between tasks based on their age-related dominance (probability of choosing a “defensive task” versus a “work task” $T \neq 0$ or 1, and $\gamma_t \neq 0$; **Table 1**). Across all scenarios, individuals are selected to perform tasks based on their likelihood of inheriting the territory (**Figure 3A**). Specifically, when subordinates prefer defensive tasks (**Figure 3A**, $m < 0.2$, circles), they engage in work tasks at a young age and increasingly undertake defensive tasks as their dominance rises (**Figure 3**, $m < 0.2$). Similarly, if their preference leans toward work tasks (**Figure 3A**, $m \geq 0.2$, circles), younger subordinate helpers and those of lower dominance focus more on defense initially, but switch to work tasks as they approach the breeding stage (**Figure 3**, $m \geq 0.2$). In essence, subordinate helpers allocate more time to tasks with lower fitness costs as they near a competitive dominance value, all without any active enforcement by other group members, resulting in age- or rank-polyethism. These results are consistent with the rise in division of labor, defined as within-individual consistency and between-individual differences in task choice [60].

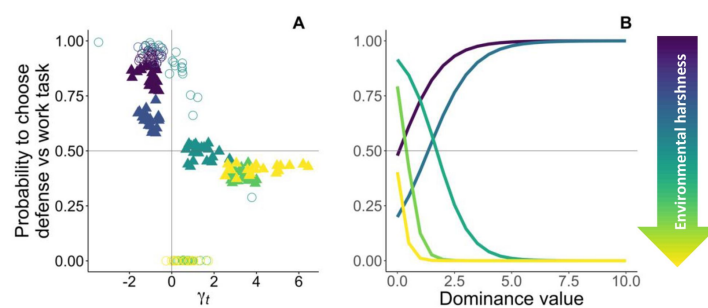


Figure 3.

Evolved reaction norms to age on the display of task specialization.

The evolutionary equilibria for the reaction norms of task specialization are shown at five different levels of environmental quality, ranging from benign ($m=0.1$, purple) to harsh ($m=0.3$, yellow), and $x_h=5$. **A:** $\gamma_t > 0$ signifies that individuals increase the likelihood of performing work tasks (task = 0; cost to dominance value), whereas $\gamma_t < 0$ signifies that individuals increase the likelihood of performing defensive tasks (task = 1; cost to survival). The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor ($\blacktriangle$), or when no restrictions were introduced to the task performed by the group members ($\circ$). Results show that at division of labor equilibria, individuals increase the likelihood of performing their preferred task (circles), when increasing their dominance value. **B:** Evolved reaction norms to dominance value with average γ_0 and γ_t across 20 replicas for varying quality environments. All parameter values described in **Table 1** [↗](#).

Discussion

Our findings suggest that direct benefits of group living play a driving role in the evolution of division of labor via task specialization in species with totipotent workers, including all vertebrates. In contrast, in eusocial insects often characterized by high relatedness and reproductive inhibition, sterile workers acquire fitness benefits only indirectly, and so, their fitness is contingent upon the overall success of the colony rather than the individual [61, 62]. Group members in eusocial species are therefore predicted to maximize colony fitness due to the associated lower within-group conflict, and to favor the evolution of division of labor to increase group productivity efficiency [48, 49, 51]. However, in taxa with lower within-group relatedness, such as nearly all vertebrates, division of labor is facilitated when cooperation is crucial for survival and when the benefits of cooperation are evenly distributed among group members [3]. In particular, survival fitness benefits derived from living in larger groups seem to be key for the evolution of cooperative behavior in vertebrates [22, 63], and may also translate into low within-group conflict. This suggests that selection for division of labor in vertebrates is stronger in smaller groups, where increased productivity significantly enhances survival, as opposed to research on social insects that posits division of labor is more likely to evolve in larger groups [64–66]. That is not to say, however, that kin selection is unimportant in vertebrates. Our model demonstrates that vertebrates evolve a more stable division of labor when direct and indirect fitness benefits act in concert. Moreover, our model predicts that the type of task that individuals prefer depends on the type of fitness benefit that subordinates can attain. While direct fitness benefits in the form of group augmentation select more strongly for defensive tasks, kin selection alone seems to select only for work tasks. In groups of unrelated individuals where members can only gain direct fitness benefits, tasks that incur costs to their dominance value and to the likelihood of attaining a breeding position are comparatively large. Conversely, groups with no group benefits tend to be small and suffer high mortality, for which defensive tasks only aggravate the situation.

Previous models of task allocation in eusocial insects have focused on how seemingly identical individuals of the same age and broad morphology may specialize in different types of tasks [8, 60]. In contrast, our model shows that extrinsic ecological factors are also important for the evolution of task specialization in addition to intrinsic factors such as age or dominance rank. In short, our model predicts that task specialization, which varies with age or rank, will depend on environmental quality. In more favorable environments, helpers are predicted to increase defensive tasks with age or rank, whereas in harsh environments, work tasks are predicted to increase with age or rank. Furthermore, harsher environments exert stronger selective pressure for task specialization due to increased survival challenges. Enhanced cooperation and improved group productivity through helping mitigate the negative effects of the environment.

This finding is consistent with previous research on the role of ecological constraints in driving the evolution of cooperation [67–69].

Although rare, evidence of division of labor in cooperative breeding vertebrates comes from a few studies in mammals [70], birds [71, 72], and fishes [73, 74]. Our model predictions are broadly consistent with empirical data from many of these studies. For example, support for the role of age and environmental harshness on the evolution of division of labor via task specialization in vertebrates comes from arguably the most similar vertebrate system to eusocial insects, the naked mole-rat (*Heterocephalus glaber*). Naked mole-rats live in self-dug burrows that protect their inhabitants from predation and climatic extremes [70]. Therefore, even though they inhabit habitats with scarce and unpredictable rainfall, environmental harshness is likely to be limited in their stable subterranean burrows, with mortality mainly spiking during dispersal attempts. Model predictions for species that evolved in benign environments are then met, with

individuals shifting from work tasks such as foraging for food, digging, and maintaining the burrow system, to defensive tasks such as guarding and patrolling as individuals grow older and larger [70, 75, 76]. Although naked mole-rats exhibit high levels of within-colony relatedness [77], group size also exerts a positive influence on survival, potentially generating group augmentation benefits [78, 79]. Under these conditions, our model predicts the highest levels of task partitioning and division of labor. In contrast, *Neolamprologus pulcher*, a cooperatively breeding cichlid fish that lives in harsh environments with high risk of predation [59], shows the opposite tendency. In line with our model predictions, larger and older helpers of this species invest relatively more in territory maintenance, whereas younger/smaller helpers defend the breeding shelter of the dominant pair to a greater extent against experimentally exposed egg predators [74]. Territory maintenance has been shown to greatly affect routine metabolic rates and, hence, growth rates [80], which directly translates into a decrease in the likelihood of becoming dominant and attaining breeding status, as predicted by our model. In addition, both group augmentation and kin selection have been found to play a role in the evolution of help in this species [20, 52]. More research is needed to verify whether this trend is widespread in cooperatively breeding vertebrates with division of labor. The preferred form of help provided by individuals throughout their life course might deviate from our expectations if helpers exhibit differing abilities to perform distinct tasks at different ages or life stages. We assumed that individuals of all ages possessed equal capability to perform different tasks. However, as individuals age and their size and body mass increase, they may become more adept at certain tasks such as deterring predators [33, 81–85]. Body condition that is unrelated to age may also influence helper contributions, as individuals in poorer condition may be more constrained on their energy allocation in activities other than self-maintenance [86–88]. Furthermore, our model does not capture instances in which help in general or specific tasks are enforced by the dominant pair or other more dominant group members [89–91].

Despite evidence of task specialization and division of labor across several vertebrate taxa, other empirical studies have failed to find similar results. For instance, researchers failed to find evidence of task specialization in Damaraland mole-rats (*Fukomys damarensis*), despite their social structure resembling that of naked mole-rats [92, 93]. However, it is worth noting that researchers have only investigated correlates between different work tasks, potentially overlooking negative correlations with other helping activities that pose a more direct cost to survival. In fact, many studies that reported an absence of division of labor did not consider defensive tasks in their analyses [92–95]. Nonetheless, studies exist in which researchers did not find a negative correlation between defense and offspring provisioning or territory maintenance [39, 96, 97]. However, individuals may exhibit a positive or non-relationship between tasks if they also vary in total helpfulness across their lifetime, but may still switch the proportion of time/effort allocated to different types of tasks. In our model, we controlled for this potential tendency by restricting individuals from changing their overall helpfulness with age/dominance rank. We encourage future researchers to consider the proportion of time/energy investment in addition to examining correlations between tasks.

In summary, our study represents the first theoretical model aimed at elucidating the potential mechanisms underlying division of labor between temporal non-reproductives via task specialization in taxa beyond eusocial organisms. Harsh environments, where individuals can obtain direct fitness benefits from group living, favor division of labor thereby enhancing group productivity and, consequently, group size. Variation in the relative fitness costs of different helping tasks with age favors temporal polyethism, with individuals changing their behavior patterns to improve their chances of becoming breeders. Hence, future empirical research on division of labor should prioritize cooperatively breeding species inhabiting relatively harsh environments where subordinates benefit greatly from group membership, such as protection or improved food acquisition. Additionally, we emphasize the importance of including sentinel and

defense behaviors in future studies to encompass a broader spectrum of fitness costs of cooperation, and to investigate potential changes in task investment throughout the helpers' lifetimes.

Materials and Methods

We developed an individual-based model in which individuals may display and vary their efforts in different helping tasks. The ancestral state features an absence of alloparental care and no task specialization. We assume that residing in larger groups yields overall survival advantages, potentially resulting in direct fitness benefits associated with the increase in group size, as proposed by the group augmentation hypothesis. We also include the coevolution of the helping strategies with their dispersal propensity, as dispersal affects group size and kin structure, and thus may impact the strength of indirect fitness benefits.

Life cycle

We model a population consisting of a fixed number of breeding territories, each comprising a dominant breeder that reproduces asexually and monopolizes reproduction, as well as an undefined number of subordinates that queue for the breeding position. The number of subordinates is determined by the productivity of the breeders and the dispersal decisions of the group members. Subordinates may help increase the fecundity of the breeder, so that the breeder's fecundity depends on the cumulative level of help provided by the subordinates within the group, with an optimum increase when different helping tasks are performed to a similar degree.

Offspring inherit the genes for dispersal and helping tendencies from the breeders, potentially exhibiting slight variation in the presence of mutations (**Figure 1** [↗](#), Step 1). Mutations occur at each genetic locus at a frequency of μ , causing minor adjustments to the allele's value inherited from the parent. This alteration involves the addition of a value drawn from a normal distribution characterized by a mean of 0 and a standard deviation of $\sigma\mu$ (**Table 1** [↗](#)).

After breeders reproduce, their offspring and the subordinates in the group may disperse independently from each other. Dispersal propensity is controlled by a gene with values ranging from 0 to 1, where 0 represents complete philopatry and 1 represents consistent dispersal (**Table 1** [↗](#)). Dispersers may migrate to another group to become subordinates or remain as floaters waiting for breeding opportunities (**Figure 1** [↗](#), Step 2). Successful dispersers enter the breeding queue within the group based on their dominance value rather than starting at the bottom of the hierarchy. Individuals that remain in a group express some level of help in the form of alloparental care, which can potentially evolve to zero (**Figure 1** [↗](#), Step 3). To be conservative, we use the term help as synonymous with altruistic behavior, which increases the fitness of the beneficiary (i.e., breeder) with an immediate decrease in the actor's fitness. Hence, cooperative behaviors such as defense from adult predators that incur immediate benefits to the actor as well as by-product mutualistic benefits to other group members, are not considered in this study. Dispersal propensity, alloparental care provisions, and the type of helping task they perform all depend on their genetic predisposition. They may perform tasks that reduce their body condition and, therefore, their chances of becoming breeders, such as feeding offspring or territory maintenance (hereafter called “work tasks”), or they may perform tasks that reduce their immediate survival such as predator defense (hereafter called “defensive tasks”) [[32](#) [↗](#)]. For simplification, we assume complete dichotomous, short-term non-cumulative fitness consequences between the different types of tasks (i.e., survival and dominance costs are only sustained during the breeding cycle in which they perform the task, and individuals only perform one of the tasks in the given cycle). We note that the names given for each type of task are meant to facilitate interpretation, and that in some species, some tasks classified as work tasks may have

a higher immediate cost in survival than in the likelihood to breed, and vice versa. Note that authors should classify helping tasks in their model species repertoire based on the primary fitness cost associated with each type of helping activity (e.g., if a work activity mostly impacts survival in a given species, this task is described in this paper as defensive).

Individual survival depends on environmental quality, group membership, group size, and the amount of help provided during defensive tasks to the breeder's offspring (**Figure 1**, Step 4). Environmental quality is expressed as a varying maximum survival rate, independent of social factors. Additional environmental effects, such as variation in mortality linked to dispersal, or on the likelihood of dispersers finding a new group to start breeding, have been explored elsewhere [63].

If the breeder in a group dies, all subordinates in the group and a sample of floaters compete for the breeding position (**Figure 1**, Step 5). The number of floaters bidding for an empty breeding spot is then proportional to the relative abundance of floaters with regard to the total number of breeding territories, to account for spatial viscosity and the likelihood of being accepted in a new group (**Table 1**). In this model, age influences the likelihood of attaining the breeding position. Older individuals, owing to their higher resource holding potential, have a greater probability of becoming breeders, as observed in various species [59, 98–106]. Age in our model corresponds to the number of breeding cycles that an individual survives. Additionally, work tasks reduce an individual's dominance value as a result of the impact on their body condition.

The likelihood of filling an empty breeding position is then implemented as a lottery weighted by the dominance value of the candidates. Therefore, subordinates may stay in the natal group and queue to inherit the dominant position [100, 107–110], but they may also disperse and breed or queue in another group [34, 108, 111].

We ran simulations for 200'000 generations or until equilibrium was reached for all genetic traits, across 20 replicas to assess consistency.

Age-dependent task specialization

The likelihood of choosing defensive versus work tasks T takes a logistic function with boundaries between 0 and 1 as given in Eq. 1, in which individuals may adjust their helping task as they age,

$$T = \frac{1}{1 + \exp(-\gamma_t t - \gamma_0)} \quad (1),$$

where γ_t modifies the strength and direction of the effect of age (t) on task choice, and γ_0 acts as the intercept. Therefore, when T approaches zero, individuals specialize in work tasks throughout their lives, and when T approaches one, they specialize in defensive tasks. If $\gamma_t \neq 0$, given that $T \neq 0$ and $T \neq 1$, individuals show age-dependent task specialization.

In addition, the phenotypic value of help (H) is regulated by another gene and equals the allelic value of gene α , which remains fixed throughout the helper's life. Even though individuals are likely to change the total amount of help given throughout their lives, we make this simplification to allow easier interpretation of how individuals may switch between task types as they age. For more details on how individuals may adapt their level of help with age and social and environmental conditions, see [63]. The fitness costs of helping are described in the following section.

Fitness costs of different helping tasks

Helping tasks are classified in our model according to their short-term fitness consequences. We assume that work tasks incur a more immediate cost to the body condition of helpers because of expenses in terms of time and energy. Work tasks reduce the dominance value R to inherit the breeding position in our model following Eq. 2 [\[112\]](#), where γ_h is a parameter that influences the strength of the cost of help (H_{work}) in the dominance value. We assume that dominance is influenced by age (t) as a proxy for resource holding potential, with older individuals having a greater dominance value,

$$R = t - \gamma_h H_{work} \quad (2).$$

Note that the cost to an individual's dominance value is influenced by their current age and the current specific task that they are performing, the cost of which is not cumulative over time. Therefore, an individual's effective rank within the dominance hierarchy is, irrespective of whether they belong to a group or not, determined by their dominance value relative to other competitors for the breeding position, with higher dominance values corresponding to higher ranks in the hierarchy.

In contrast, we assume that defensive tasks have a more immediate fitness cost on survival (e.g., predator or space competitor defense) due to the high risk of injury. Therefore, helpers that defend have a survival rate (S_H) proportional to the amount of help provided ($H_{defence}$) given by the logistic Eq. 3.1 [\[112\]](#), where x_h is a parameter that influences the strength of the cost of help in survival,

$$S_H = \frac{1-m}{1+\exp(-x_0+x_h H_{defence}-x_h N)} \quad (3.1).$$

In addition to defensive tasks, social and ecological factors also influence the survival of group members. The term m denotes the baseline mortality that determines how harsh the environment is. Higher values of m indicate higher overall environmental mortality for all individuals in the population, irrespective of social factors. For simplicity, we assume that the harshness of the environment only impacts survival, although other effects such as on reproductive output or the strength in which help improves fecundity are likely to have an effect in nature [\[112\]](#). Group size (N) provides survival benefits for both subordinates and breeders (Eq. 3.1 [\[112\]](#) and 3.2 [\[112\]](#) respectively), where x_h is a parameter that quantifies the effect size of the benefit of group size in survival, and x_0 an intercept. Floaters sustain the highest mortality (S_F , Eq. 3.3 [\[112\]](#)) since they do not benefit from group protection,

$$S_B = \frac{1-m}{1+\exp(-x_0-x_h N)} \quad (3.2) \text{ and}$$

$$S_F = \frac{1-m}{1+\exp(-x_0)} \quad (3.3).$$

Need for division of labor

To assess the rules governing task specialization and division of labor, we first outline a basic model in which individuals evolve their preferred helping task and then compare it to a scenario in which the breeder's reproductive outcome is maximized by performing both tasks to an approximately equivalent extent. Thus, the maximum amount of cumulative help of each type (ΣH_{max}) that can affect fecundity is given by Eq. 4 [\[112\]](#):

$$\Sigma H_{max} = \frac{\Sigma H_{defence} + \Sigma H_{work}}{2} + k_m \quad (4),$$

where k_m is a parameter that relaxes the need for division of labor. The breeders then reproduced asexually. The quantity of offspring produced is contingent upon the breeders' baseline fecundity (k_0) and the collective amount of help contributed by the helpers in the group. The increase in fecundity shows diminishing returns as given in the Eq. 5 [\[113\]](#), where k_h is a scaling parameter that modifies the benefit of the cumulative help provided by the subordinates to increase the fecundity of the breeder,

$$K = k_0 + \frac{k_h(\sum H_{defence} + \sum H_{work})}{1 + \sum H_{defence} + \sum H_{work}} \quad (5).$$

If $\sum H_{defence} > \sum H_{max}$ or $\sum H_{work} > \sum H_{max}$, then we use the value H_{max} instead. The breeder's fecundity is determined by a randomly selected value drawn from a Poisson distribution with mean of K .

Fitness benefits of helping

We include both direct and indirect fitness benefits of helping to assess the relative importance of each in the evolution of task specialization. Indirect fitness benefits from relatedness among group members (i.e., kin selection) emerge from demographic factors such as dispersal tendencies, mortality rates, and group dynamics, though they were not coded explicitly. To calculate the coefficient of relatedness between the breeder and subordinates in a group, we calculated the coefficient of a linear regression between the allelic values of the breeders and helpers of a neutral gene that changes exclusively by genetic drift [\[113\]](#), [\[114\]](#). Direct benefits may also be obtained by enhancing group size by helping to increase the breeder's fecundity, as larger groups benefit from higher survival rates in our model (i.e., group augmentation).

To distinguish the effect of kin selection from group augmentation, we created two parallel models for comparison to the default model (group augmentation and kin selection acting independently), in which one of the fitness benefits was removed. In the "only group augmentation" implementation, individuals just born (i.e., age = 1) that decide to stay in the natal territory as subordinates (**Figure 1** [\[115\]](#), Step 2) are shuffled into another group, thereby removing the influence of relatedness from the model. In the "only kin selection" implementation, we removed the survival benefits linked to living in a group ($x_n = 0$). It is possible that direct and indirect fitness benefits could have an additive effect on the evolution of alloparental care, since group size enhancement has a positive effect on the survival of kin group members, not only on the individual.

Supporting information

Kin selection and the evolution of division of labor

To assess the robustness of our findings on the evolution of division of labor driven by kin selection, independent of group survival benefits, we further examined the influence of survival, dispersal, relatedness, and breeding opportunities outside the natal group.

One factor contributing to defense not readily evolving under kin selection alone is that the cost to dominance is relatively small when competing against relatives. Another possible explanation is that without group benefits, mortality rates are too high for survival-costly helping tasks such as defense to evolve. In **Figure S1** [\[116\]](#), we examine the effects of increasing the baseline survival from $x_0 = 1.5$ (as in **Figure 2** [\[117\]](#)) to $x_0 = 4.5$ (which simulates the addition of a group benefit of $x_n = 3$ to the baseline survival) and $x_0 = 10$ (an extreme scenario that greatly reduces the survival cost of defense tasks; see Eq. 3.1 [\[118\]](#)).

Our results indicate that even with high baseline survival, kin selection strongly favors work over defense tasks. However, in harsh environments ($m = 0.3$), where the survival costs of defense tasks are minimal ($x_0 = 10$ and $x_h = 3$; **Table S2** [↗](#)), some degree of task specialization and division of labor emerges (**Figure S1** [↗](#)). Despite this, most individuals still choose to disperse and breed outside their natal territory when no group survival benefits exist (**Table S2** [↗](#)). This suggests that, without such benefits, offspring have little incentive to stay and help due to reduced reproductive opportunities and competition with kin. To test this, we eliminated the dispersal advantage by reducing the number of sampled groups from two ($f = 2$) to one ($f = 1$), incentivizing individuals to queue for breeding in their natal territory (**Figure S2** [↗](#)).

Even when increased philopatry, resulting from reduced outbreeding opportunities for dispersers, potentially enhances the indirect fitness benefits of helping and task partitioning, division of labor only evolves when survival costs are exceedingly low ($x_0 = 10$; **Figure S2** [↗](#); **Table S3** [↗](#)). This suggests that while kin selection can promote the evolution of division of labor, it does so only under highly restrictive conditions with minimal survival costs. Therefore, in natural settings, other selective forces, such as direct fitness benefits, are more likely to drive the evolution of division of labor.

Acknowledgements

We thank Patrick Kennedy for comments on earlier versions of the manuscript and Tobias Zobrist for help with the coding implementation.

Additional information

Funding

Funding was provided by the SNSF grant P500PB_214371 to Irene García-Ruiz.

Author Contributions

conceptualization IGR & DR, methodology IGR, investigation IGR, visualization IGR, supervision DR, writing—original draft IGR, writing—review & editing DR.

Data and materials availability

Custom code used to generate the simulations is available at https://github.com/IreneGR92/Task_specialization [↗](#)

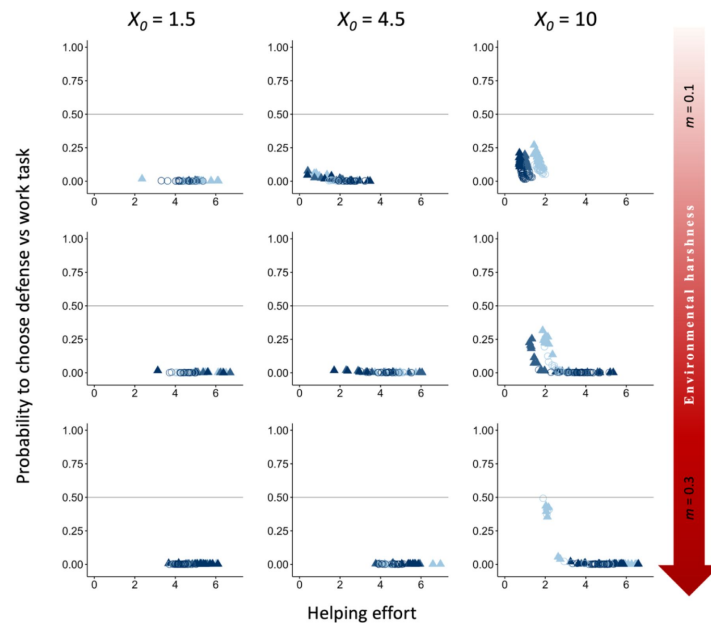


Figure S1.

Effect of increasing the baseline survival x_0 to favor the evolution of division of labor under only kin selection.

The evolutionary equilibria for levels of helping and task specialization are shown at three different levels of environmental quality, ranging from benign ($m = 0.1$) to harsh ($m = 0.3$), and for three different levels of cost of help on survival (light blue, $x_h = 3$; environmental = 5; and, ranging $x_h = 7$). The y-axis expresses the likelihood of individuals choosing a defensive task with a cost to survival versus a work task with a cost to dominance. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor ($\blacktriangle$), or when no restrictions were introduced to the task performed by the group members ($\circ$). Input parameters are the same as in **Figure 2** (where $x_0 = 1.5$) except for $x_0 = 4.5$ and $x_0 = 10$ (higher survival for all individuals irrespective of group membership or environment; **Table 1**). Further details are provided in **Table S2**.

Figure S2.

Effect of reducing the incentives to disperse to favor the evolution of division of labor under only kin selection.

An increase in incentives to remain philopatric was achieved by reducing f to 1 ($f = 2$ in Figure 2; Table 1). The evolutionary equilibria for levels of helping and task specialization are shown at three different levels of environment quality that range from benign ($m=0.1$) to harsh ($m = 0.3$), and for three different levels of cost of help on survival (light blue, $x_h = 3$; blue, $x_h = 5$; and dark blue, $x_h = 7$). The y-axis expresses the likelihood of the individuals to choose a defensive task with cost to survival versus a work task with a cost to dominance. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor (▲), or when no restrictions were introduced to the task performed by the group members (○). Other input parameters are the same as in Figure 2 (where $x_0 = 1.5$) except for $x_0 = 4.5$ and $x_0 = 10$ (higher survival for all individuals irrespective of group membership or environment; Table 1).

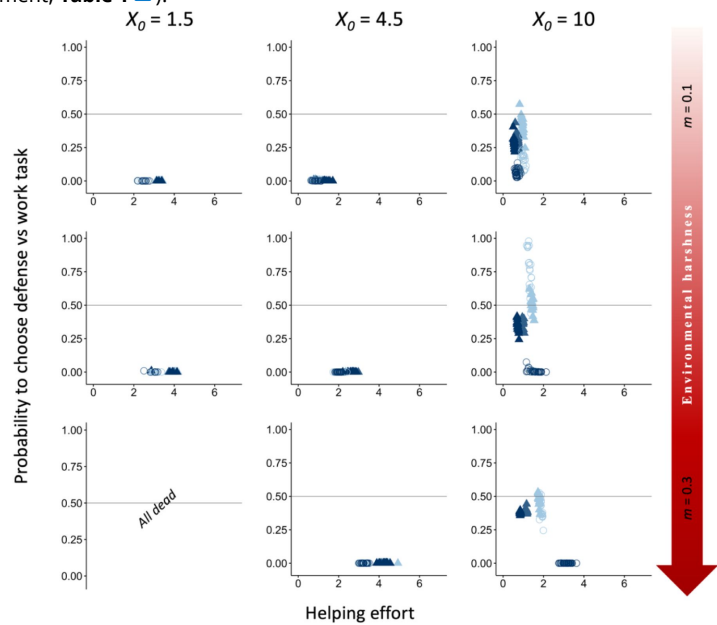


Table S1.

Supplementary data for Figure 2.

Mean values are shown for dispersal propensity, survival rate, group size ($\pm$ SD), and within-group relatedness for three environmental qualities ranging from benign ($m = 0.1$) to harsh ($m = 0.3$) across 20 replicas. Selective forces at play include kin selection (KS), group augmentation (GA), or both (KS + GA). The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor (DoL), or when no restrictions were introduced to the task performed by the group members (No DoL).

	KS						GA						KS + GA					
	$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$	
	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL
Dispersal	0.95	0.92	0.94	0.90	0.91	0.84	0.01	0.01	0.11	0.10	0.37	0.33	0.07	0.07	0.22	0.22	0.43	0.52
Survival	0.74	0.74	0.65	0.65	0.57	0.57	0.90	0.90	0.79	0.79	0.66	0.67	0.89	0.89	0.77	0.77	0.65	0.65
Group size	1.16 ± 0.16	1.34 ± 0.30	1.14 ± 0.12	1.25 ± 0.19	1.14 ± 0.08	1.29 ± 0.11	9.85 ± 0.08	9.83 ± 0.05	6.94 ± 0.43	7.52 ± 0.41	3.04 ± 0.35	3.40 ± 0.36	16.42 ± 0.10	16.55 ± 0.07	6.42 ± 0.14	6.61 ± 0.11	2.95 ± 0.25	2.83 ± 0.05
Relatedness	0.37	0.36	0.54	0.52	0.73	0.71	0	0	0	0	0	0	0.56	0.57	0.57	0.57	0.65	0.60

Table S2.

Supplementary data for the effect of increasing the baseline survival x_0 to favor the evolution of division of labor under only kin selection shown in Figure S1.

Mean values are shown for dispersal propensity, survival rate, group size ($\pm$ SD), and within-group relatedness for three environmental qualities ranging from benign ($m = 0.1$) to harsh ($m = 0.3$) across 20 replicas. Results are shown for $x_0 = 1.5$ (default), $x_0 = 3.5$ and $x_0 = 10$. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor (DoL), or when no restrictions were introduced to the task performed by the group members (No DoL).

	$x_0 = 1.5$						$x_0 = 4.5$						$x_0 = 10$					
	$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$	
	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL
Dispersal	0.95	0.92	0.94	0.9	0.91	0.84	0.84	0.85	0.89	0.86	0.89	0.85	0.79	0.83	0.88	0.85	0.9	0.86
Survival	0.74	0.74	0.65	0.65	0.57	0.57	0.89	0.89	0.79	0.79	0.69	0.69	0.90	0.90	0.80	0.80	0.70	0.70
Group size	1.16 ± 0.16	1.34 ± 0.30	1.14 ± 0.12	1.25 ± 0.19	1.14 ± 0.08	1.29 ± 0.11	3.02 ± 0.33	3.14 ± 0.14	1.61 ± 0.26	1.86 ± 0.21	1.33 ± 0.13	1.5 ± 0.09	4.19 ± 0.19	3.75 ± 0.20	1.68 ± 0.22	1.96 ± 0.07	1.28 ± 0.11	1.48 ± 0.07
Relatedness	0.37	0.36	0.54	0.52	0.73	0.71	0.15	0.15	0.28	0.28	0.45	0.44	0.17	0.15	0.27	0.27	0.44	0.43

Table S3.

Supplementary data for the effect of reducing the incentives to disperse to favor the evolution of division of labor under only kin selection shown in Figure S2.

The increased incentive was achieved by reducing the parameter f that signifies the mean number of groups a floater samples for becoming a breeder from 2 (default) to 1. Mean values are shown for dispersal propensity, survival rate, group size ($\pm$ SD), and within-group relatedness for three environmental qualities ranging from benign ($m = 0.1$) to harsh ($m = 0.3$) across 20 replicas. Results are shown for $x_0 = 1.5$ (default), $x_0 = 3.5$ and $x_0 = 10$. The optimum breeder productivity per unit of help provided was either when both tasks were performed to a similar extent, potentially selecting for division of labor (DoL), or when no restrictions were introduced to the task performed by the group members (No DoL).

	$x_0 = 1.5$						$x_0 = 4.5$						$x_0 = 10$					
	$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$		$m = 0.1$		$m = 0.2$		$m = 0.3$	
	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL	DoL	No DoL
Dispersal	0.82	0.75	0.79	0.79	NA	NA	0.23	0.24	0.49	0.50	0.72	0.68	0.25	0.25	0.44	0.47	0.49	0.66
Survival	0.74	0.74	0.65	0.65	0	0	0.89	0.89	0.79	0.79	0.69	0.69	0.90	0.90	0.80	0.80	0.70	0.70
Group size	1.90 ± 1.07	2.34 ± 1.26	1.71 ± 0.62	1.75 ± 0.75	0	0	13.09 ± 0.23	13.35 ± 0.40	4.84 ± 0.38	4.90 ± 0.08	2.04 ± 0.06	2.33 ± 0.05	14.29 ± 0.26	14.59 ± 0.16	5.17 ± 0.37	5.30 ± 0.08	3.10 ± 0.26	2.47 ± 0.05
Relatedness	0.40	0.39	0.55	0.56	NA	NA	0.32	0.32	0.37	0.36	0.46	0.46	0.29	0.29	0.38	0.36	0.52	0.46

References

1. Dunbar RIM, Lehmann J, Korstjens AJ, Gowlett JAJ (2014) **The road to modern humans: time budgets, fission-fusion sociality, kinship and the division of labour in hominin evolution** In: Dunbar RIM, Gamble C, Gowlett JAJ, editors. *Lucy to language: The benchmark papers* Oxford University Press Oxford pp. 333–355
2. Smith A (1776) **An Inquiry into the Nature and Causes of the Wealth of Nations** London: Wiley Online Library
3. Cooper GA, West SA (2018) **Division of labour and the evolution of extreme specialization** *Nat Ecol Evol* **2**:1161–1167 <https://doi.org/10.1038/s41559-018-0564-9>
4. Zhang Z, Claessen D, Rozen DE (2016) **Understanding Microbial Divisions of Labor** *Front Microbiol* **7**:2070 <https://doi.org/10.3389/fmicb.2016.02070>
5. Oster GF, Wilson EO (1978) **Caste and ecology in the social insects** Princeton University Press
6. Jeanne RL (2016) **Division of labor is not a process or a misleading concept** *Behav Ecol Sociobiol* **70**:1109–1112 <https://doi.org/10.1007/s00265-016-2146-7>
7. Michener CD (1974) **The social behavior of the bees: a comparative study** Harvard University Press
8. Beshers SN, Fewell JH (2001) **Models of division of labor in social insects** *Annu Rev Entomol* **46**:413–440 <https://doi.org/10.1146/annurev.ento.46.1.413>
9. Wilson EO (1971) **The insect societies**. Cambridge, Massachusetts USA: Harvard University Press
10. Wheeler WM (1928) **Emergent evolution and the development of societies** WW Norton, Incorporated
11. Ispolatov I, Ackermann M, Doebeli M (2012) **Division of labour and the evolution of multicellularity** *Proceedings of the Royal Society B: Biological Sciences* **279**:1768–1776 <https://doi.org/10.1098/rspb.2011.1999>
12. Willensdorfer M (2009) **On the evolution of differentiated multicellularity** *Evolution* **63**:306–323 <https://doi.org/10.1111/j.1558-5646.2008.00541.x>
13. Yanni D, Jacobeen S, Márquez-Zacarias P, et al (2020) **Topological constraints in early multicellularity favor reproductive division of labor** *eLife* **9**:e54348 <https://doi.org/10.7554/eLife.54348>
14. Ridley AR, Raihani NJ (2008) **Task partitioning increases reproductive output in a cooperative bird** *Behavioral Ecology* **19**:1136–1142 <https://doi.org/10.1093/beheco/arn097>
15. Dunn PO, Cockburn A, Mulder RA (1995) **Fairy-wren helpers often care for young to which they are unrelated** *Proc R Soc Lond B Biol Sci* **259**:339–343 <https://doi.org/10.1098/rspb.1995.0050>

16. Doutrelant C, Dalecky A, Covas R (2011) **Age and relatedness have an interactive effect on the feeding behaviour of helpers in cooperatively breeding sociable weavers** *Behaviour* **148**:1393–1411 <https://doi.org/10.1163/000579511X608701>
17. Riehl C (2013) **Evolutionary routes to non-kin cooperative breeding in birds** *Proceedings of the Royal Society B: Biological Sciences* **280**:20132245 <https://doi.org/10.1098/rspb.2013.2245>
18. Kokko H, Johnstone RA, Clutton-Brock TH (2001) **The evolution of cooperative breeding through group augmentation** *Proceedings of the Royal Society B: Biological Sciences* **268**:187–196 <https://doi.org/10.1098/rspb.2000.1349>
19. Kingma SA, Santema P, Taborsky M, Komdeur J (2014) **Group augmentation and the evolution of cooperation** *Trends Ecol Evol* **29**:476–484 <https://doi.org/10.1016/j.tree.2014.05.013>
20. Zöttl M, Heg D, Chervet N, Taborsky M (2013) **Kinship reduces alloparental care in cooperative cichlids where helpers pay-to-stay** *Nat Commun* **4**:1341 <https://doi.org/10.1038/ncomms2344>
21. Sparkman AM, Adams JR, Steury TD, et al (2011) **Direct fitness benefits of delayed dispersal in the cooperatively breeding red wolf (*Canis rufus*)** *Behavioral Ecology* **22**:199–205 <https://doi.org/10.1093/beheco/arq194>
22. Shah SS, Rubenstein DR (2023) **Group augmentation underlies the evolution of complex sociality in the face of environmental instability** *Proceedings of the National Academy of Sciences* **120**:e2212211120 <https://doi.org/10.1073/pnas.2212211120>
23. Richardson DS, Burke T, Komdeur J (2002) **Direct benefits and the evolution of female-biased cooperative breeding in Seychelles warblers** *Evolution* **56**:2313–2321 <https://doi.org/10.1111/j.0014-3820.2002.tb00154.x>
24. Cockburn A (1998) **Evolution of Helping Behavior in Cooperatively Breeding Birds** *Annu Rev Ecol Syst* **29**:141–177 <https://doi.org/10.1146/annurev.ecolsys.29.1.141>
25. Kitchen D, Beehner J (2007) **Factors affecting individual participation in group-level aggression among non-human primates** *Behaviour* **144**:1551–1581 <https://doi.org/10.1163/156853907782512074>
26. Anderson C (2001) **Teams in animal societies** *Behavioral Ecology* **12**:534–540 <https://doi.org/10.1093/beheco/12.5.534>
27. Nunn Charles L., Lewis Rebecca J. (2001) **Cooperation and collective action in animal behaviour** In: Noë Ronald, Van Hooff Jan A.R.A.M., Hammerstein Peter, editors. *Economics in Nature: Social Dilemmas, Mate Choice and Biological Markets* Cambridge (United Kingdom): Cambridge University Press, pp. 42–66
28. Poiani A, Yorke M (1989) **Predator Harassment: More Evidence on the Deadly Risk** *Ethology* **83**:167–169 <https://doi.org/10.1111/j.1439-0310.1989.tb00526.x>
29. Tórrez L, Robles N, González A, Crofoot MC (2012) **Risky business? Lethal attack by a jaguar sheds light on the costs of predator mobbing for capuchins (*Cebus capucinus*)** *Int J Primatol* **33**:440–446 <https://doi.org/10.1007/s10764-012-9588-1>

30. Sordahl TA (1990) **The risks of avian mobbing and distraction behavior: an anecdotal review** *Wilson Bull* **102**:349–352
31. Grantner A, Taborsky M (1998) **The metabolic rates associated with resting, and with the performance of agonistic, submissive and digging behaviours in the cichlid fish *Neolamprologus pulcher* (Pisces: Cichlidae)** *J Comp Physiol B* **168**:427–433 <https://doi.org/10.1007/s003600050162>
32. Heinsohn R, Legge S (1999) **The cost of helping** *Trends Ecol Evol* **14**:53–57 [https://doi.org/10.1016/S0169-5347\(98\)01545-6](https://doi.org/10.1016/S0169-5347(98)01545-6)
33. Heinsohn R, Cockburn A (1994) **Helping is costly to young birds in cooperatively breeding white-winged choughs** *Proceedings of the Royal Society B: Biological Sciences* **256**:293–298 <https://doi.org/10.1098/rspb.1994.0083>
34. Zöttl M, Chapuis L, Freiburghaus M, Taborsky M (2013) **Strategic reduction of help before dispersal in a cooperative breeder** *Biol Lett* **9**:20120878 <https://doi.org/10.1098/rsbl.2012.0878>
35. Young AJ, Carlson AA, Clutton-Brock T (2005) **Trade-offs between extraterritorial prospecting and helping in a cooperative mammal** *Anim Behav* **70**:829–837 <https://doi.org/10.1016/j.anbehav.2005.01.019>
36. Komdeur J (1996) **Influence of age on reproductive performance in the Seychelles warbler** *Behavioral Ecology* **7**:417–425 <https://doi.org/10.1093/beheco/7.4.417>
37. Boland CRJ, Heinsohn R, Cockburn A (1997) **Deception by helpers in cooperatively breeding white-winged choughs and its experimental manipulation** *Behav Ecol Sociobiol* **41**:251–256 <https://doi.org/10.1007/s002650050386>
38. Taborsky M (1984) **Broodcare helpers in the cichlid fish *Lamprologus brichardi*: Their costs and benefits** *Anim Behav* **32**:1236–1252 [https://doi.org/10.1016/S0003-3472\(84\)80241-9](https://doi.org/10.1016/S0003-3472(84)80241-9)
39. Clutton-Brock TH, Russell AF, Sharpe LL (2003) **Meerkat helpers do not specialize in particular activities** *Anim Behav* **66**:531–540 <https://doi.org/10.1006/anbe.2003.2209>
40. Boesch C (2002) **Cooperative hunting roles among taï chimpanzees** *Human Nature* **13**:27–46 <https://doi.org/10.1007/s12110-002-1013-6>
41. Fischer B, van Doorn GS, Dieckmann U, Taborsky B (2014) **The evolution of age- dependent plasticity** *American Naturalist* **183**:108–125 <https://doi.org/10.1086/674008>
42. Johnstone RA, Cant MA (2010) **The evolution of menopause in cetaceans and humans: the role of demography** *Proceedings of the Royal Society B: Biological Sciences* **277**:3765–3771 <https://doi.org/10.1098/rspb.2010.0988>
43. Hölldobler B, Wilson EO (1990) **The ants** Harvard University Press
44. Sendova-Franks A, Franks NR (1993) **Task allocation in ant colonies within variable environments (a study of temporal polyethism: experimental)** *Bull Math Biol* **55**:75–96 <https://doi.org/10.1007/BF02460295>
45. Traniello JFA, Rosengaus RB (1997) **Ecology, evolution and division of labour in social insects** *Anim Behav* **53**:209–213 <https://doi.org/10.1006/anbe.1996.0289>

46. Ferguson-Gow H, Sumner S, Bourke AFG, Jones KE (2014) **Colony size predicts division of labour in attine ants** *Proceedings of the Royal Society B: Biological Sciences* **281**:20141411 <https://doi.org/10.1098/rspb.2014.1411>
47. Keller L, Chapuisat M (2010) **Eusociality and Cooperation** In: *Encyclopedia of Life Sciences* Wiley
48. West SA, Fisher RM, Gardner A, Kiers ET (2015) **Major evolutionary transitions in individuality** *Proceedings of the National Academy of Sciences* **112**:10112–10119 <https://doi.org/10.1073/pnas.1421402112>
49. Fisher RM, Cornwallis CK, West SA (2013) **Group formation, relatedness, and the evolution of multicellularity** *Current Biology* **23**:1120–1125
50. Queller DC, Strassmann JE (1998) **Kin selection and social insects** *Bioscience* **48**:165–175
51. Gardner A, Grafen A (2009) **Capturing the superorganism: a formal theory of group adaptation** *J Evol Biol* **22**:659–671 <https://doi.org/10.1111/j.1420-9101.2008.01681.x>
52. García-Ruiz I, Taborsky M (2022) **Group augmentation on trial: helpers in small groups enhance antipredator defence of eggs** *Biol Lett* **18**:20220170 <https://doi.org/10.1098/rsbl.2022.0170>
53. Rood JP (1990) **Group size, survival, reproduction, and routes to breeding in dwarf mongooses** *Anim Behav* **39**:566–572 [https://doi.org/10.1016/S0003-3472\(05\)80423-3](https://doi.org/10.1016/S0003-3472(05)80423-3)
54. Heg D, Brouwer L, Bachar Z, Taborsky M (2005) **Large group size yields group stability in the cooperatively breeding cichlid *Neolamprologus pulcher*** *Behaviour* **142**:1615–1641 <https://doi.org/10.1163/156853905774831891>
55. Clutton-Brock TH, Gaynor D, McIlrath GM, et al (1999) **Predation, group size and mortality in a cooperative mongoose, *Suricata suricatta*** *Journal of Animal Ecology* **68**:672–683 <https://doi.org/10.1046/j.1365-2656.1999.00317.x>
56. Guindre-Parker S, Rubenstein DR (2020) **Survival Benefits of Group Living in a Fluctuating Environment** *Am Nat* **195**:1027–1036 <https://doi.org/10.1086/708496>
57. Bilde T, Coates KS, Birkhofer K, et al (2007) **Survival benefits select for group living in a social spider despite reproductive costs** *J Evol Biol* **20**:2412–2426 <https://doi.org/10.1111/j.1420-9101.2007.01407.x>
58. Rubenstein DR, Koenig WD (2016) **Superb starlings: cooperation and conflict in an unpredictable environment** In: Koenig WD, Dickinson JL, editors. *Cooperative breeding in vertebrates: studies of ecology, evolution, and behavior* Cambridge: Cambridge University Press pp. 181–196
59. Taborsky M (2016) **Cichlid fishes: A model for the integrative study of social behavior** In: Koenig WD, Dickinson JL, editors. *Cooperative Breeding in Vertebrates* Cambridge: Cambridge University Press pp. 272–293
60. Ulrich Y, Saragosti J, Tokita CK, et al (2018) **Fitness benefits and emergent division of labour at the onset of group living** *Nature* **560**:635–638 <https://doi.org/10.1038/s41586-018-0422-6>
61. Hughes WOH, Oldroyd BP, Beekman M, Ratnieks FLW (2008) **Ancestral monogamy shows kin selection is key to the evolution of eusociality** *Science* **320**:1213–1216 <https://doi.org/10>

.1126/science.1156108

62. Foster K, Wenseleers T, Ratnieks F (2006) **Kin selection is the key to altruism** *Trends Ecol Evol* **21**:57–60 <https://doi.org/10.1016/j.tree.2005.11.020>
63. García-Ruiz I, Quiñones A, Taborsky M (2022) **The evolution of cooperative breeding by direct and indirect fitness effects** *Sci Adv* **8**:eabl7853 <https://doi.org/10.1126/sciadv.abl7853>
64. Nakahashi W, Feldman MW (2014) **Evolution of division of labor: Emergence of different activities among group members** *J Theor Biol* **348**:65–79 <https://doi.org/10.1016/j.jtbi.2014.01.027>
65. Dornhaus A, Powell S, Bengston S (2012) **Group Size and Its Effects on Collective Organization** *Annu Rev Entomol* **57**:123–141 <https://doi.org/10.1146/annurev-ento-120710-100604>
66. Ulrich Y, Saragosti J, Tokita CK, et al (2018) **Fitness benefits and emergent division of labour at the onset of group living** *Nature* **560**:635–638 <https://doi.org/10.1038/s41586-018-0422-6>
67. Hatchwell BJ, Komdeur J (2000) **Ecological constraints, life history traits and the evolution of cooperative breeding** *Anim Behav* **59**:1079–1086 <https://doi.org/10.1006/anbe.2000.1394>
68. Emlen ST (1982) **The evolution of helping. I. An ecological constraints model** *The American Naturalist* **119**:29–39 <https://doi.org/10.1086/283888>
69. Kokko H, Ekman J (2002) **Delayed dispersal as a route to breeding: Territorial inheritance, safe havens, and ecological constraints** *American Naturalist* **160**:468–484 <https://doi.org/10.1086/342074>
70. Lacey EA, Sherman PW (1996) **Cooperative breeding in naked mole-rats: implications for vertebrate and invertebrate sociality** In: Solomon NG, French JA, editors. *Cooperative Breeding in Mammals* Cambridge University Press pp. 267–301
71. Arnold KE, Owens IPF, Goldizen AW (2005) **Division of labour within cooperatively breeding groups** *Behaviour* **142**:1577–1590 <https://doi.org/10.1163/156853905774831927>
72. Ridley AR, Raihani NJ (2008) **Task partitioning increases reproductive output in a cooperative bird** *Behavioral Ecology* **19**:1136–1142 <https://doi.org/10.1093/beheco/arn097>
73. Josi D, Freudiger A, Taborsky M, Frommen JG (2021) **Experimental predator intrusions in a cooperative breeder reveal threat-dependent task partitioning** *Behavioral Ecology* **31**:1369–1378 <https://doi.org/10.1093/beheco/araa094>
74. Bruintjes R, Taborsky M (2011) **Size-dependent task specialization in a cooperative cichlid in response to experimental variation of demand** *Anim Behav* **81**:387–394 <https://doi.org/10.1016/j.anbehav.2010.10.004>
75. Lacey EA, Sherman PW (1991) **Social organization of naked mole-rat colonies: evidence for divisions of labor** In: Sherman PW, Jarvis JUM, Alexander RD, editors. *The Biology of the Naked Mole-Rat* Princeton University Press pp. 275–336
76. Faulkes CG, Abbott DH, Jarvis JUM (1991) **Social suppression of reproduction in male naked mole-rats, *Heterocephalus glaber*** *Reproduction* **91**:593–604 <https://doi.org/10.1530/jrf.0.0910593>

77. Reeve HK, Westneat DF, Noon WA, et al (1990) **DNA “fingerprinting” reveals high levels of inbreeding in colonies of the eusocial naked mole-rat** *Proceedings of the National Academy of Sciences* **87**:2496–2500 <https://doi.org/10.1073/pnas.87.7.2496>
78. Le Comber SC, Spinks AC, Bennett NC, et al (2002) **Fractal dimension of African mole-rat burrows** *Can J Zool* **80**:436–441 <https://doi.org/10.1139/z02-026>
79. O’Riain MJ, Braude S (2001) **Inbreeding versus outbreeding in captive and wild populations of naked mole-rats** In: Clobert J, Danchin E, Dhondt AA, Nichols JD, editors. *Dispersal* Oxford University Press Oxford pp. 143–154
80. Taborsky M, Grantner A (1998) **Behavioural time–energy budgets of cooperatively breeding *Neolamprologus pulcher* (Pisces: Cichlidae)** *Anim Behav* **56**:1375–1382 <https://doi.org/10.1006/anbe.1998.0918>
81. Brown JL (1985) **The evolution of helping behavior—an ontogenetic and comparative perspective** In: Gollin ES, editors. *The comparative development of adaptive skills* pp. 137–171
82. Clutton-Brock TH, Russell AF, Sharpe LL, et al (2002) **Evolution and development of sex differences in cooperative behavior in meerkats** *Science* **297**:253–256 <https://doi.org/10.1126/science.1071412>
83. Erb WM, Porter LM (2020) **Variable infant care contributions in cooperatively breeding groups of wild saddleback tamarins** *Am J Primatol* **82**:e23190 <https://doi.org/10.1002/ajp.23190>
84. Zahed SR, Kurian A V., Snowdon CT (2010) **Social dynamics and individual plasticity of infant care behavior in cooperatively breeding cotton-top tamarins** *Am J Primatol* **72**:296–306 <https://doi.org/10.1002/ajp.20782>
85. Woxvold IA, Mulder RA, Magrath MJL (2006) **Contributions to care vary with age, sex, breeding status and group size in the cooperatively breeding apostlebird** *Anim Behav* **72**:63–73 <https://doi.org/10.1016/j.anbehav.2005.08.016>
86. Clutton-Brock TH, Brotherton PNM, O’Riain MJ, et al (2000) **Individual contributions to babysitting in a cooperative mongoose** *Suricata suricatta*. *Proceedings of the Royal Society B: Biological Sciences* **267**:301–305 <https://doi.org/10.1098/rspb.2000.1000>
87. Clutton-Brock TH, Brotherton PNM, O’Riain MJ, et al (2001) **Contributions to cooperative rearing in meerkats** *Anim Behav* **61**:705–710 <https://doi.org/10.1006/anbe.2000.1631>
88. Nichols HJ, Amos W, Bell MBV, et al (2012) **Food availability shapes patterns of helping effort in a cooperative mongoose** *Anim Behav* **83**:1377–1385 <https://doi.org/10.1016/j.anbehav.2012.03.005>
89. García-Ruiz I, Taborsky M (2024) **Nepotism mediates enforced cooperation in asymmetric negotiations** *iScience* **27**:110334 <https://doi.org/10.1016/j.isci.2024.110334>
90. Kokko H, Johnstone RA, Wright J (2002) **The evolution of parental and alloparental effort in cooperatively breeding groups: when should helpers pay to stay?** *Behavioral Ecology* **13**:291–300 <https://doi.org/10.1093/beheco/13.3.291>
91. Hellmann JK, Hamilton IM (2018) **Dominant and subordinate outside options alter help and eviction in a pay-to-stay negotiation model** *Behavioral Ecology* **29**:553–562 <https://doi.org>

[/10.1093/beheco/ary006](https://doi.org/10.1093/beheco/ary006)

92. Zöttl M, Vullioud P, Mendonça R, et al (2016) **Differences in cooperative behavior among Damaraland mole rats are consequences of an age-related polyethism** *Proceedings of the National Academy of Sciences* **113**:10382–10387 <https://doi.org/10.1073/pnas.1607885113>
93. Thorley J, Mendonça R, Vullioud P, et al (2018) **No task specialization among helpers in Damaraland mole-rats** *Anim Behav* **143**:9–24 <https://doi.org/10.1016/j.anbehav.2018.07.004>
94. English S, Nakagawa S, Clutton-Brock TH (2010) **Consistent individual differences in cooperative behaviour in meerkats (*Suricata suricatta*)** *J Evol Biol* **23**:1597–1604
95. Smith MG, LaPergola JB, Riehl C (2024) **Workload inequality increases with group size in a cooperatively breeding bird** *Anim Behav* **207**:87–99 <https://doi.org/10.1016/j.anbehav.2023.10.015>
96. van Asten T, Hall ML, Mulder RA (2016) **Who cares? Effect of coping style and social context on brood care and defense in superb fairy-wrens** *Behavioral Ecology* **27**:1745–1755 <https://doi.org/10.1093/beheco/arw096>
97. Teunissen N, Kingma SA, Peters A (2020) **Nest defence and offspring provisioning in a cooperative bird: individual subordinates vary in total contribution, but no division of tasks among breeders and subordinates** *Behav Ecol Sociobiol* **74**:94 <https://doi.org/10.1007/s00265-020-02877-2>
98. Kingma SA, Hall ML, Peters A (2011) **Multiple benefits drive helping behavior in a cooperatively breeding bird: An integrated analysis** *American Naturalist* **177**:486–495 <https://doi.org/10.1086/658989>
99. Shaw P (1986) **The relationship between dominance behaviour, bill size and age group in Greater Sheathbills *Chionis alba*** *Ibis* **128**:48–56 <https://doi.org/10.1111/j.1474-919X.1986.tb02091.x>
100. Bridge C, Field J (2007) **Queuing for dominance: Gerontocracy and queue-jumping in the hover wasp *Liostenogaster flavolineata*** *Behav Ecol Sociobiol* **61**:1253–1259 <https://doi.org/10.1007/s00265-007-0355-9>
101. Doolan SP, Macdonald DW (1996) **Dispersal and extra-territorial prospecting by slender-tailed meerkats (*Suricata suricatta*) in the south-western Kalahari** *J Zool* **240**:59–73 <https://doi.org/10.1111/j.1469-7998.1996.tb05486.x>
102. Parker GA (1974) **Assessment strategy and the evolution of fighting behaviour** *J Theor Biol* **47**:223–243 [https://doi.org/10.1016/0022-5193\(74\)90111-8](https://doi.org/10.1016/0022-5193(74)90111-8)
103. Brown JL, Brown ER, Sedransk J (1997) **Dominance, age, and reproductive success in a complex society: a long-term study of the mexican jay** *Auk* **114**:279–286 <https://doi.org/10.2307/4089168>
104. Buston PM (2004) **Territory inheritance in clownfish** *Proc R Soc Lond B Biol Sci* **271**:S252–S254 <https://doi.org/10.1098/rsbl.2003.0156>
105. Bang A, Gadagkar R (2012) **Reproductive queue without overt conflict in the primitively eusocial wasp *Ropalidia marginata*** *Proceedings of the National Academy of Sciences* **109**:14494–14499 <https://doi.org/10.1073/pnas.1212698109>

106. Unnikrishnan S, Gadagkar R (2021) **Dominance behaviour and division of labour in the tropical primitively eusocial wasp *Ropalidia cyathiformis*** *Insectes Soc* **68**:123–132 <https://doi.org/10.1007/s00040-020-00803-3>
107. Woolfenden GE, Fitzpatrick JW (1978) **The inheritance of territory in group-breeding birds** *Bioscience* **28**:104–108 <https://doi.org/10.2307/1307423>
108. Russell EM, Rowley I (1993) **Philopatry or dispersal: competition for territory vacancies in the splendid fairy-wren, *Malurus splendens*** *Anim Behav* **45**:519–539 <https://doi.org/10.1006/anbe.1993.1063>
109. Dierkes P, Heg D, Taborsky M, et al (2005) **Genetic relatedness in groups is sex- specific and declines with age of helpers in a cooperatively breeding cichlid** *Ecol Lett* **8**:968–975 <https://doi.org/10.1111/j.1461-0248.2005.00801.x>
110. Balshine-Earn S, Neat FC, Reid H, Taborsky M (1998) **Paying to stay or paying to breed? Field evidence for direct benefits of helping behavior in a cooperatively breeding fish** *Behavioral Ecology* **9**:432–438 <https://doi.org/10.1093/beheco/9.5.432>
111. Walters JR, Doerr PD, Carter JH (1988) **The cooperative breeding system of the red-cockaded woodpecker** *Ethology* **78**:275–305 <https://doi.org/10.1111/j.1439-0310.1988.tb00239.x>
112. Rubenstein DR (2011) **Spatiotemporal environmental variation, risk aversion, and the evolution of cooperative breeding as a bet-hedging strategy** *Proceedings of the National Academy of Sciences* **108**:10816–10822
113. Gardner A, West SA, Wild G (2011) **The genetical theory of kin selection** *J Evol Biol* **24**:1020–1043 <https://doi.org/10.1111/j.1420-9101.2011.02236.x>
114. Michod RE, Hamilton WD (1980) **Coefficients of relatedness in sociobiology** *Nature* **288**:694–697 <https://doi.org/10.1038/288694a0>

Author information

Irene García-Ruiz

Department of Ecology, Evolution and Environmental Biology, Columbia University, New York, United States

ORCID iD: [0000-0002-6139-5857](https://orcid.org/0000-0002-6139-5857)

For correspondence: igaru.13@gmail.com

Dustin Rubenstein

Department of Ecology, Evolution and Environmental Biology, Columbia University, New York, United States

ORCID iD: [0000-0002-4999-3723](https://orcid.org/0000-0002-4999-3723)

Editors

Reviewing Editor

Sergio Rasmann

University of Neuchâtel, Neuchâtel, Switzerland

Senior Editor

Sergio Rasmann

University of Neuchâtel, Neuchâtel, Switzerland

Reviewer #1 (Public review):

This paper presents a computational model of the evolution of two different kinds of helping ("work," presumably denoting provisioning, and defense tasks) in a model inspired by cooperatively breeding vertebrates. The helpers in this model are a mix of previous offspring of the breeder and floaters that might have joined the group, and can either transition between the tasks as they age or not. The two types of help have differential costs: "work" reduces "dominance value," (DV), a measure of competitiveness for breeding spots, which otherwise goes up linearly with age, but defense reduces survival probability. Both eventually might preclude the helper from becoming a breeder and reproducing. How much the helpers help, and which tasks (and whether they transition or not), as well as their propensity to disperse, are all evolving quantities. The authors consider three main scenarios: one where relatedness emerges from the model, but there is no benefit to living in groups, one where there is no relatedness, but living in larger groups gives a survival benefit (group augmentation, GA), and one where both effects operate. The main claim is that evolving defensive help or division of labor requires the group augmentation; it doesn't evolve through kin selection alone in the authors' simulations.

This is an interesting model, and there is much to like about the complexity that is built in. Individual-based simulations like this can be a valuable tool to explore the complex interaction of life history and social traits. Yet, models like this also have to take care of both being very clear on their construction and exploring how some of the ancillary but potentially consequential assumptions affect the results, including robust exploration of the parameter space. I think the current manuscript falls short in these areas, and therefore, I am not yet convinced of the results. Much of this is a matter of clearer and more complete writing: the Materials and Methods section in particular is incomplete or vague in some important junctions. However, there are also some issues with the assumptions that are described clearly.

Below, I describe my main issues, mostly having to do with model features that are unclear, poorly motivated (as they stand), or potentially unrealistic or underexplored.

One of the main issues I have is that there is almost no information on what happens to dispersers in the model. Line 369-67 states dispersers might join another group or remain as floaters, but gives no further information on how this is determined. Poring through the notation table also comes up empty as there is no apparent parameter affecting this consequential life history event. At some point, I convinced myself that dispersers remain floaters until they die or become breeders, but several points in the text contradict this directly (e.g., l 107). Clearly this is a hugely important model feature since it determines fitness cost and benefits of dispersal and group size (which also affects relatedness and/or fitness depending on the model). There just isn't enough information to understand this crucial component of the model, and without it, it is hard to make sense of the model output.

Related to that, it seems to be implied (but never stated explicitly) that floaters do no work, and therefore their DV increases linearly with age (H_{work} in eq.2 is zero). That means any floaters that manage to stick around long enough would have higher success in competition for breeding spots relative to existing group members. How realistic is this? I think this might be driving the kin selection-only results that defense doesn't evolve without group augmentation (one of the two main ways). Any subordinates (which are mainly zero in the no GA, according to the SI tables; this assumes $N = \text{breeder} + \text{subordinates}$, but this isn't explicit anywhere) would be outcompeted by floaters after a short time (since they evolve high H and floaters don't), which in turn increases the benefit of dispersal, explaining why it is so high. Is this parameter regime reasonable? My understanding is that floaters often aren't usually high resource holding potential individuals (either b/c high RHP ones would get selected out of the floater population by establishing territories or b/c floating isn't typically a thriving strategy, given that many resources are tied to territories). In this case, the assumption seems to bias things towards the floaters and against subordinates to inherit territories. This should be explored either with a higher mortality rate for floaters and/or a lower DV increase, or both.

When it comes to floaters replacing dead breeders, the authors say a bit more, but again, the actual equation for the scramble competition (which only appears as "scramble context" in the notation table) is not given. Is it simply proportional to $R_i / \sum_j R_j$? Or is there some other function used? What are the actual numbers of floaters per breeding territory that emerge under different parameter values? These are all very important quantities that have to be described clearly.

I also think the asexual reproduction with small mutations assumption is a fairly strong one that also seems to bias the model outcomes in a particular way. I appreciate that the authors actually measured relatedness within groups (though if most groups under KS have no subordinates, that relatedness becomes a bit moot), and also eliminated it with their ingenious swapping-out-subordinates procedure. The fact remains that unless they eliminate relatedness completely, average relatedness, by design, will be very high. (Again, this is also affected by how the fate of the dispersers is determined, but clearly there isn't a lot of joining happening, just judging from mean group sizes under KS only.) This is, of course, why there is so much helping evolving (even if it's not defensive) unless they completely cut out relatedness.

Finally, the "need for division of labor" section is also unclear, and its construction also would seem to bias things against division of labor evolving. For starters, I don't understand the rationale for the convoluted way the authors create an incentive for division of labor. Why not implement something much simpler, like a law of minimum (i.e., the total effect of helping is whatever the help amount for the lowest value task is) or more intuitively: the fecundity is simply a function of "work" help (draw Poisson number of offspring) and survival of offspring (draw binomial from the fecundity) is a function of the "defense" help. As it is, even though the authors say they require division of labor, in fact, they only make a single type of help marginally less beneficial (basically by half) if it is done more than the other. That's a fairly weak selection for division of labor, and to me it seems hard to justify. I suspect either of the alternative assumptions above would actually impose enough selection to make division of labor evolve even without group augmentation.

Overall, this is an interesting model, but the simulation is not adequately described or explored to have confidence in the main conclusions yet. Better exposition and more exploration of alternative assumptions and parameter space are needed.

<https://doi.org/10.7554/eLife.105501.1.sa2>

Reviewer #2 (Public review):**Summary:**

This paper formulates an individual-based model to understand the evolution of division of labor in vertebrates. A main conclusion of the paper is that direct fitness benefits are the primary factor causing the evolution of vertebrate division of labor, rather than indirect fitness benefits.

Strengths:

The paper formulates an individual-based model that is inspired by vertebrate life history. The model incorporates numerous biologically realistic details, including the possibility to evolve age polytheism where individuals switch from work to defence tasks as they age or vice versa, as well as the possibility of comparing the action of group augmentation alone with that of kin selection alone.

Weaknesses:

The model makes assumptions that restrict the possibility that kin selection leads to the evolution of helping. In particular, the model assumes that in the absence of group augmentation, subordinates can only help breeders but cannot help non-breeders or increase the survival of breeders, whereas with group augmentation, subordinates can help both breeders and non-breeders and increase the survival of breeders. This is unrealistic as subordinates in real organisms can help other subordinates and increase the survival of non-breeders, even in the absence of group augmentation, for instance, with targeted helping to dominants or allies. This restriction artificially limits the ability of kin selection alone to lead to the evolution of helping, and potentially to division of labor. Hence, the conclusion that group augmentation is the primary driving factor driving vertebrate division of labor appears forced by the imposed restrictions on kin selection. The model used is also quite particular, and so the claimed generality across vertebrates is not warranted.

I describe some suggestions for improving the paper below, more or less in the paper's order.

First, the introduction goes to great lengths trying to convince the reader that this model is the first in this or another way, particularly in being only for vertebrates, as illustrated in the abstract where it is stated that "we lack a theoretical framework to explore the conditions under which division of labor is likely to evolve" (line 13). However, this is a risky and unnecessary motivation. There are many models of division of labor and some of them are likely to be abstract enough to apply to vertebrates even if they are not tailored to vertebrates, so the claims for being first are not only likely to be wrong but will put many readers in an antagonistic position right from the start, which will make it harder to communicate the results. Instead of claiming to be the first or that there is a lack of theoretical frameworks for vertebrate division of labor, I think it is enough and sufficiently interesting to say that the paper formulates an individual-based model motivated by the life history of vertebrates to understand the evolution of vertebrate division of labor. You could then describe the life history properties that the model incorporates (subordinates can become reproductive, low relatedness, age polyethism, etc.) without saying this has never been done or that it is exclusive to vertebrates; indeed, the paper states that these features do not occur in eusocial insects, which is surprising as some "primitively" eusocial insects show them. So, in short, I think the introduction should be extensively revised to avoid claims of being the first and to make it focused on the question being addressed and how it is addressed. I think this could be done in 2-3 paragraphs without the rather extensive review of the literature in the current introduction.

Second, the description of the model and results should be clarified substantially. I will give specific suggestions later, but for now, I will just say that it is unclear what the figures show. First, it is unclear what the axes in Figure 2 show, particularly for the vertical one. According to the text in the figure axis, it presumably refers to T , but T is a function of age t , so it is unclear what is being plotted. The legend explaining the triangle and circle symbols is unintelligible (lines 227-230), so again it is unclear what is being plotted; part of the reason for this unintelligibility is that the procedure that presumably underlies it (section starting on line 493) is poorly explained and not understandable (I detail why below). Second, the axes in Figure 3 are similarly unclear. The text in the vertical axis in panel A suggests this is T , however, T is a function of t and γ_t , so something else must be being done to plot this. Similarly, in panel B, the horizontal axis is presumably R , but R is a function of t and of the helping genotype, so again some explanation is lacking. In all figures, the symbol of what is being plotted should be included.

Third, the conclusions sound stronger than the results are. A main conclusion of the paper is that "kin selection alone is unlikely to select for the evolution of defensive tasks and division of labor in vertebrates" (lines 194-195). This conclusion is drawn from the left column in Figure 2, where only kin selection is at play, and the helping that evolves only involves work rather than defense tasks. This conclusion follows because the model assumes that without group augmentation (i.e., $x_n=0$, the kin selection scenario), subordinates can only help breeders to reproduce but cannot help breeders or other subordinates to survive, so the only form of help that evolves is the least costly, not the most beneficial as there is no difference in the benefits given among forms of helping. This assumption is unrealistic, particularly for vertebrates where subordinates can help other group members survive even in the absence of group augmentation (e.g., with targeted help to certain group members, because of dominance hierarchies where the helping would go to the breeder, or because of alliances where the helping would go to other subordinates). I go into further details below, but in short, the model forces a narrow scope for the kin selection scenario, and then the paper concludes that kin selection alone is unlikely to be of relevance for the evolution of vertebrate division of labor. This conclusion is particular to the model used, and it is misleading to suggest that this is a general feature of such a particular model.

Overall, I think the paper should be revised extensively to clarify its aims, model, results, and scope of its conclusions.

<https://doi.org/10.7554/eLife.105501.1.sa1>

Author response:

We will revise the statements of novelty in the introduction by more clearly emphasizing how our model addresses gaps in the existing literature. In addition, we will clarify the description of the dispersal process. Briefly, we use the same dispersal gene β to represent the likelihood an individual will either leave or join a group, thereby quantifying both dispersal and immigration using the same parameter. Specifically, individuals with higher β are more likely to remain as floaters (i.e., disperse from their natal group to become a breeder elsewhere), whereas those with lower β are either more likely to remain in their natal group as subordinates (i.e., queue in a group for the breeding position) or join another group if they dispersed. Immigrants that join a group as a subordinate help and queue for a breeding position, as does any natal subordinate born into the group. To follow the suggestion of the referee and more fully explore the impact of competition between subordinates born in the group and subordinate immigrants, we will explore extending our model to allow dispersers to leave their natal group and join another as subordinates, by incorporating a reaction norm based on their age or rank ($D = 1 / (1 + \exp(\beta_t * t - \beta_0))$). This approach will allow individuals

to adjust also their dispersal strategy to their competitiveness and to avoid kin competition by remaining as a subordinate in another group.

We apologize that there was some confusion with terminology. We use the term “disperser” to describe individuals that disperse from their natal group. Dispersers can assume one of three roles: (1) they can migrate to another group as “subordinates”; (2) they can join another group as “breeders” if they successfully outcompete other candidates; or (3) they can remain as “floaters” if they fail to join a group. “Floaters” are individuals who persist in a transient state without access to a breeding territory, waiting for opportunities to join a group in an established territory. Therefore, dispersers do not work when they are floaters, but they may later help if they immigrate to a group as a subordinate. Consequently, immigrant subordinates have no inherent competitive advantage over natal subordinates (as step 2.2. “Join a group” is followed by step 3. “Help”, which occurs before step 5. “Become a breeder”). Nevertheless, floaters can potentially outcompete subordinates of the same age if they attempt to breed without first queuing as a subordinate (step 5) when subordinates are engaged in work tasks. We believe that this assumption is realistic and constitutes part of the costs associated with work tasks. However, floaters are at a disadvantage for becoming a breeder because: (1) floaters incur higher mortality than individuals within groups (eq. 3); and (2) floaters may only attempt to become breeders in some breeding cycles (versus subordinate group members, who are automatically candidates for an open breeding position in the group in each cycle). Therefore, due to their higher mortality, floaters are rarely older than individuals within groups, which heavily influences dominance value and competitiveness. Additionally, any competitive advantage that floaters might have over other subordinate group members is unlikely to drive the kin selection-only results because subordinates would preferably choose defense tasks instead of work tasks so as not to be at a competitive disadvantage compared to floaters.

We note that reviewers also mention that floaters often aren't usually high resource holding potential (RHP) individuals and, therefore, our assumptions might be unrealistic. As we explain above, floaters are not inherently at a competitive advantage in our model. In any case, empirical work in a number of species has shown that dispersers are not necessarily those of lower RHP or of lower quality. In fact, according to the ecological constraints hypothesis, one might predict that high quality individuals are the ones that disperse because only individuals in good condition (e.g., larger body size, better energy reserves) can afford the costs associated with dispersal (Cote et al., 2022). By adding a reaction norm approach to explore the role of age or rank in the revised version, we can also determine whether higher or lower quality individuals are the ones dispersing. We will address the issues of terminology and clarity of the relative competitive advantage of floaters versus subordinates, and also include more information in the Supplementary Tables (e.g., the number of floaters). As a side note, the “scramble context” we mention was an additional implementation that we decided to remove from the final manuscript, but we forgot to remove from Table 1 before submission.

The reviewers also raised a question about asexual reproduction and relatedness more generally. As we showed in the Supplementary Tables and the section on relatedness in the SI (“Kin selection and the evolution of division of labor”), high relatedness does not appear to explain our results. In evolutionary biology generally and in game theory specifically (with the exception of models on sexual selection or sex-specific traits), asexual reproduction is often modelled because it reduces unnecessary complexity. To further study the effect of relatedness on kin structures more closely resembling those of vertebrates, however, we will create an additional “relatedness structure level”, where we will shuffle half of the philopatric offspring using the same method used to remove relatedness completely. This approach will effectively reduce relatedness structure by half and overcome the concerns with our decision to model asexual reproduction.

Briefly, we will elaborate on the concept of division of labor and the tasks that cooperative breeders perform. In nature, multiple tasks are often necessary to successfully rear offspring. For example, in many cooperatively breeding birds, the primary reasons that individuals fail to produce offspring are (1) starvation, which is mitigated by the feeding of offspring, and (2) nest depredation, which is countered by defensive behavior. Consequently, both types of tasks are necessary to successfully produce offspring, and focusing solely on one while neglecting the other is likely to result in lower reproductive success than if both tasks are performed by individuals within the group. We simplify this principle in the model by maximizing reproductive output when both tasks are carried out to a similar extent, allowing for some flexibility from the mean. In response to the reviewer suggestion about making fecundity a function of work tasks and offspring survival as a function of defensive tasks, these are actually equivalent in model terms, as it's the same whether breeders produce three offspring and two die, or if they only produce one. This represents, of course, a simplification of the natural context, where breeding unsuccessfully is more costly (in terms of time and energy investment) than not breeding at all, but this is approach is typically used in models of this sort.

The scope of this paper was to study division of labor in cooperatively breeding species with fertile workers, in which help is exclusively directed towards breeders to enhance offspring production (i.e., alloparental care). Our focus is in line with previous work in most other social animals, including eusocial insects and humans, which emphasizes how division of labor maximizes group productivity. Other forms of “general” help are not considered in the paper, and such forms of help are rarely considered in cooperatively breeding vertebrates or in the division of labor literature, as they do not result in task partitioning to enhance productivity.

How do we model help? Help provided is an interaction between H (total effort) and T (proportion of total effort invested in each type of task). We will make this definition clearer in the revised manuscript. Thank you for pointing out an error in Eq. 1. This inequality was indeed written incorrectly in the paper (but is correct in the model code); it is dominance rank instead of age (see code in `Individual.cpp` lines 99-119). We will correct this mistake in the revision.

There was also a question about bounded and unbounded helping costs. The difference in costs is inherent to the nature of the different task (work or defense): while survival is naturally bounded, with death as the lower bound, dominance costs are potentially unbounded, as they are influenced by dynamic social contexts and potential competitors. Therefore, we believe that the model's cost structure is not too different to that in nature.

Thank you for your comments about the parameter landscape. It is important to point out that variations in the mutation rate do not qualitatively affect our results, as this is something we explored in previous versions of the model (not shown). Briefly, we find that variations in the mutation rates only alter the time required to reach equilibrium. Increasing the step size of mutation diminishes the strength of selection by adding stochasticity and reducing the genetic correlation between offspring and their parents. Population size could, in theory, affect our results, as small populations are more prone to extinction. Since this was not something we planned to explore in the paper directly, we specifically chose a large population size, or better said, a large number of territories (i.e. 5000) that can potentially host a large population.

During the exploratory phase of the model development, various parameters and values were also assessed. However, the manuscript only details the ranges of values and parameters where changes in the behaviors of interest were observed, enhancing clarity and conciseness. For instance, variation in y_h (the cost of help on dominance when performing “work tasks”) led to behavioral changes similar to those caused by changes in x_h (the cost of

help in survival when performing “defensive tasks”), as both are proportional to each other. Specifically, since an increase in defense costs raises the proportion of work relative to defense tasks, while an increase in the costs of work task has the opposite effect, only results for the variation of x_h were included in the manuscript to avoid redundancy. We will make this clearer in the revision.

Finally, following the advice from the reviewers, we will add the symbols of the variables to the figure axes, and clarify whether the values shown represent a genetic or phenotypic trait. In Figure 2, the x-axis is H and the y-axis is T . In Figure 3A, the subindex t in x-axis is incorrect; it should be subindex R (reaction norm to dominance rank instead of age), the y-axis is T . In Figure 3B, the x-axis is R , and the y-axis is T . All values of T , H and R are phenotypic expressed values (see Table 1). For instance, T values are the phenotypic expressed values from the individuals in the population according to their genetic gamma values and their current dominance rank at a given time point.

References

Cote, J., Dahirel, M., Schtickzelle, N., Altermatt, F., Ansart, A., Blanchet, S., Chaine, A. S., De Laender, F., De Raedt, J., & Haegeman, B. (2022). Dispersal syndromes in challenging environments: A cross-species experiment. *Ecology Letters*, 25(12), 2675–2687.

<https://doi.org/10.7554/eLife.105501.1.sa0>